

Assist-as-needed Ankle Rehabilitation based on Adaptive Impedance Control

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Abstract—This paper presents an assist-as-needed method for ankle rehabilitation based on adaptive impedance control strategies. First, the dynamic contribution of the patient during the motion is estimated from robot's torque and kinematic information. Then, two robot assistance control strategies (complementary and optimal) are proposed to calculate the stiffness parameter of the impedance controller. In addition, the robot level of assistance is adapted based on patient's performance in a video game. The proposed strategies were implemented using the Anklebot and evaluated in four post-stroke patients for dorsi/plantarflexion and inversion/eversion movements. Results indicate that the patient's stiffness estimation is a valid method to determine the amount of the assistance and to help the patient to complete the tasks. Finally, the results confirm that increasing the participation and performance of the patient generates a decrease in the robotic assistance, and vice versa.

I. INTRODUCTION

Stroke is the leading cause of permanent disabilities in the world. The expected prevalence of stroke and other neurological diseases will affect 350 million people by 2030, close to 5% of the world population [1]. Presently the incidence of stroke is over 15 million new cases each year with 50 million stroke survivors. More than 50% of the survivors are left permanently disabled and in need of physical therapy to promote additional gains and prevent further decline.

Robot-assisted rehabilitation has shown a positive impact and in some cases better results than conventional therapy (American Heart Guidelines). However, these results are achieved only if the robotic therapy is part of a comprehensive rehabilitation process managed by a physical therapist [2] and promotes the active participation of the patient [3]. One promising approach in rehabilitation robotics is to assist the patient only when s/he cannot complete the therapy unassisted. These systems modify the amount of assistance according to the online measurement of patient's performance [4][5][6].

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Impedance control [7] is a common approach in assist-as-needed robotic rehabilitation systems to adjust the amount of robotic assistance. The robot does not assist, promoting active participation of the patient, unless the patient deviates from the desired motion [8]. Furthermore, the real-time variation of the impedance control parameters (inertia, damping and stiffness) can determine the level of assistance offered by the robot. Use of impedance-controlled devices that modify their parameters according to different control objectives, as adaptation or optimization, may improve the assistance-as-needed rehabilitation systems [4].

In this paper, we investigate two adaptive control strategies that aim to assist only as needed and promote the patient's participation during the therapy session. These strategies modify the amount of robotic assistance based on the real-time estimation of the dynamical behavior of the patient-robot system. To validate the method and demonstrate patient's participation, we present here an error-based patient's stiffness estimation. Two methods to compute the necessary robot assistance are proposed, resulting in adaptive impedance control strategies in which the robot's stiffness is adjusted. A set of serious games is used to evoke the movements and monitor the patient's performance. The level of the robotic assistance is also adapted according to patient's performance in the games. Complementing the results presented in [9], in this paper both dorsi/plantarflexion and inversion/eversion movements are considered during the test of the proposed methodology in four post-stroke patients using an ankle rehabilitation robot [10].

The paper is organized as follows: Section II presents the description of the robotic device and the games, the adaptive impedance control strategies, and the experimental protocol used to evaluate it; Section III presents the experimental results; Section IV presents a discussion about the results, and Section V presents the conclusions.

II. METHODS

A. Experimental Setup

To evaluate the proposed methodology, we used the wearable, light and backdriveable ankle robot Anklebot (Interactive Motion Technologies, Watertown, MA, USA). The Anklebot [10] acts on the ankle joint via two linear actuators mounted in parallel to the leg. Actuated movements are dorsi/plantarflexion (DP), when the actuators move in



Fig. 1. Anklebot.

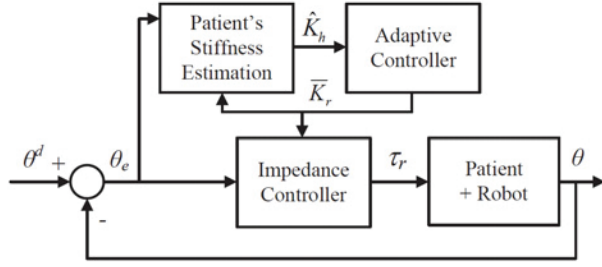


Fig. 2. Adaptive Impedance Control.

the same direction, and inversion/eversion (IE), when the actuators move in opposite directions, Figure 1.

The Anklebot employs a simple PD impedance controller with a controllable torsional stiffness and damping, and programmable reference positions. In order to make an adaptive impedance control, we calculate online the stiffness parameter of the controller from the estimated patient's stiffness (based on the position error), Figure 2.

We used two visually-evoked games for seated position to guide the patient to perform dorsi/plantarflexion and inversion/eversion movements. Games consist of achieving a sequence of targets that appear alternately on two different levels of the screen (vertical to DP and horizontal to IE), Figure 3.

A constant force in the opposite direction of the desired movement is also considered. Thus, the robot is programmed to perform the dual function of assisting the patient's movements while imposing a challenge to him. The main goal of the game is to reach the largest number of targets.

B. Estimation of Patient's Ankle Stiffness

The following dynamic model of the patient-robot system is considered:

$$\tau_r + \tau_h + \tau_e = I(\theta) \ddot{\theta} + C(\theta, \dot{\theta}) + G(\theta), \quad (1)$$

where θ is the angular position (DP or IE), $I(\theta)$ is the combined patient-robot inertia, $C(\theta, \dot{\theta})$ represents the velocity-dependent effects and $G(\theta)$ represents the gravitational effects, τ_r is the robot's torque, τ_h is the patient's torque, τ_e is the simulated constant external torque.

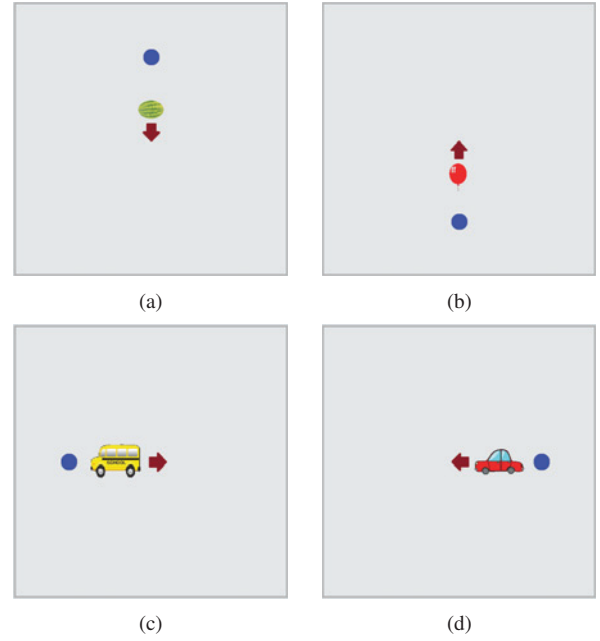


Fig. 3. Interactive environment. (a) Downward force, dorsiflexion movement; (b) Upward force, plantarflexion movement; (c-d) Sideward forces, inversion and eversion movements. Arrows represent the resistive forces (they do not appear during the game) and the blue circle is the target.

The torque generated by the robot's impedance control is:

$$\tau_r = K_r \theta_e - B_r \dot{\theta}, \quad (2)$$

where θ_e is the position error, and K_r and B_r are, respectively, the stiffness and damping of the impedance controller.

Assuming that the patient adopts an impedance control behavior similar to the robot controller, we can model the patient's torque as:

$$\tau_h = K_h \theta_e - B_h \dot{\theta}, \quad (3)$$

where K_h and B_h are, respectively, the stiffness and damping of the patient.

Replacing (2) and (3) in (1) and assuming steady state condition, the patient's stiffness can be estimated as:

$$\hat{K}_h = \frac{\tau_e}{\theta_e} - K_r, \quad (4)$$

where $\hat{K}_h$ assumes only positive values.

C. Adaptive Control Strategies

Two methods to compute the robot stiffness based on the patient's stiffness estimate are proposed [9].

1) *Complementary Adaptive Control*: We define an admissible error, θ_e^{adm} , and compute the necessary combined stiffness, $K_h + K_r$, to achieve such error as:

$$K^{adm} = \frac{\tau_e}{\theta_e^{adm}}. \quad (5)$$

The complementary robot stiffness is determined by:

$$K_r^c = K^{adm} - \hat{K}_h, \quad (6)$$

where K_r^c assumes only positive values.

2) *Optimal Adaptive Control*: According to [11], we define a cost function that characterizes the patient-robot interaction by weighting the patient error and the robot effort, as follows:

$$J = \theta_e^2 + \beta \tau_r^2, \quad (7)$$

where $\beta > 0$ is a weighting parameter.

Assuming steady state condition, isolating θ_e from (4) and substituting in (7), the final form of J is given by:

$$J = \tau_p^2 (1 + \beta K_r^2) / (K_r + \hat{K}_h)^2. \quad (8)$$

The value of K_r that optimizes the cost function is:

$$K_r^* = \frac{1}{\beta \hat{K}_h}. \quad (9)$$

In order to establish an upper limit for the robot stiffness, the value of K_r is restricted to be always smaller than K_r^c :

$$K_r^o = \min \left\{ K^{adm} - \hat{K}_h, \frac{1}{\beta \hat{K}_h} \right\}. \quad (10)$$

3) *Performance-based Adaptation*: According to (6) and (10), if the patient does not participate during a given target appearance ($\hat{K}_h \rightarrow 0$), the robot will increase its stiffness to its maximum allowed value. To promote the patient participation, we define an assistance factor that limits the robot action. The practical robot stiffness, $\bar{K}_r$, is given by:

$$\bar{K}_r = \alpha K_r, \quad (11)$$

where $0 < \alpha < 1$ is the assistance factor.

The value of α is updated by a performance-based adaptive strategy of the form:

$$\alpha_{k+1} = f\alpha_k + g(1 - P), \quad (12)$$

where k is an index for the given set of target appearances (game section), f and g are, respectively, the forgetting and gain factors, and P is the performance measurement given by the percentage of targets achieved during a given section.

D. Experimental Protocol

1) *Subjects*: The adaptive control strategies were tested in chronic stroke patients, recruited by the Department of Physical Therapy at the Federal University of São Carlos, Brazil, for voluntary participation in an assessment session. The group consists of four male patients, chronic post-stroke, three ischemic and one hemorrhagic, aged between 57 and 69 years (61.8 ± 5.5 years old), with their first event occurred between 19 and 139 months ago. All of them without significant limitation of range of motion in the ankle. Three of them have right hemiparesis and dominance. The four patients had low values for walking speed (0.55 ± 0.27 m/s), high values in the TUG (Timed Up and Go) modified test (24 ± 9.3 s) and two of them use cane bracing to walk.

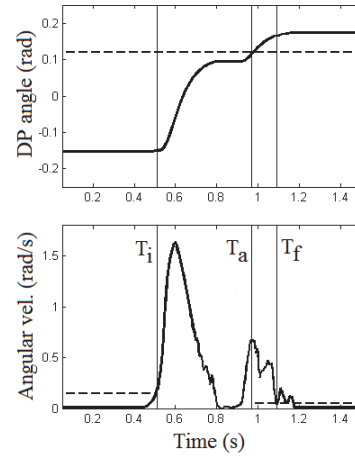


Fig. 4. Time markers to the calculation of the kinematic parameters. (Top) Angular position of the ankle during a DP movement (solid) and desired position (dashed) for this movement. (Bottom) Absolute value of the angular velocity (solid) and velocity thresholds (dashed) for calculation of time markers. T_i is the first time when the absolute value of the angular velocity exceeds 0.15 rad/s. T_a is the time when the patient reaches the desired position. T_f is the first time after reaching the target when the absolute value of the angular velocity is < 0.05 rad/s.

2) *Procedure*: Patients perform two evaluation sessions in a single day. Each session consists of 4 trials, combining the two adaptive method, complementary and optimal, and the two degrees of freedom, DP and IE. In each trial, patients perform 100 alternating movements (50 dorsiflexions/inversions and 50 plantarflexions/eversions). Patients rested for 1 min. between trials and 5 min. between each session.

Regarding the game features, the external torque was defined as 3 N.m for DP and 1.5 N.m for IE. The external torque changes gradually from negative value (in dorsiflexion/inversion) to positive value (in plantarflexion/eversion), and vice-versa. The available time was defined as 3 s for DP and 5 s for IE. The desired positions for the targets, θ_d , are defined as $\pm 7^\circ$ rad for both dof. The admissible error is defined as 20% of the desired position, i.e., $\theta_e^{adm} = 1.4^\circ$. From (5), the necessary combined stiffnesses are 125 Nm/rad for DP and 62.5 N.m/rad for IE. The assistance factor starts with $\alpha = 0.5$ and its updates every 10 movements, defined as a game sub-section. The following parameters are used: $f = 0.95$, $g = 0.1$, $B_r = 1$ Nms/rad, and $\beta = 0.0005$. This study was approved by the Ethics Committee of the Federal University of São Carlos (Number 26054813.1.0000.5504)

E. Kinematic Parameters

In order to analyze the proposed strategies based only on kinematic and kinetic data, we calculated a set of parameters for each movement successfully performed, i.e., when the target is successfully reached. As first step, we identified time markers that define relevant phases during the movements, Figure 4. The following parameters are defined:

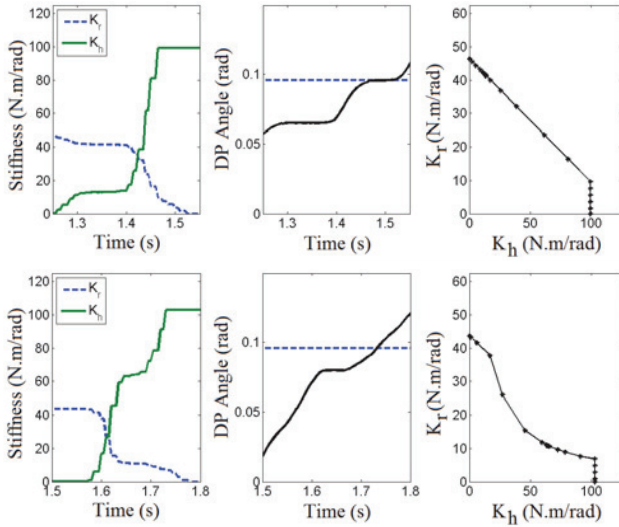


Fig. 5. Typical responses for (left) estimated patient's stiffness (green) and robot's stiffness (blue); (middle) ankle DP position (continuous), and desired position (dashed); and (right) robot's stiffness vs. estimated patient's stiffness for (top) complementary and (bottom) optimal adaptive control.

- 1) The time required for complete the movement:

$$T_r = T_a - T_i. \quad (13)$$

- 2) The mean angular velocity during the movement:

$$\omega_m = \frac{1}{T_f - T_i} \int_{T_i}^{T_f} |\omega(t)| dt. \quad (14)$$

- 3) The mean square error between T_i and T_a :

$$\theta_e^{RMS} = \sqrt{\frac{1}{T_a - T_i} \int_{T_i}^{T_a} (\theta_e(t))^2 dt}. \quad (15)$$

- 4) The robotic assistance during the movement:

$$A_R = \int_{T_i}^{T_a} K_r(t) \theta_e(t) dt. \quad (16)$$

- 5) The total cost value for each movement:

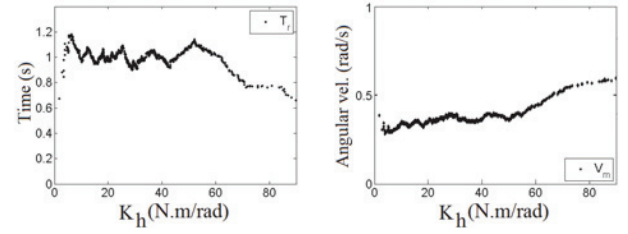
$$J_T = \int_{T_i}^{T_a} (\theta_e^2(t) + \beta \tau_r^2(t)) dt. \quad (17)$$

III. RESULTS

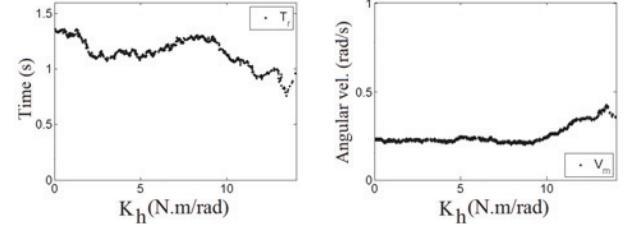
A. Typical Response for $\hat{K}_h$ e K_r

We modified online the robotic assistance according to the estimated participation of the patient. Typical behavior for the estimation process of the patient's stiffness and calculation of robot stiffness is shown in Figure 5. The figure shows a successful movement and only the time period when the patient's stiffness estimate is not zero.

Estimation and assistance algorithms start 0.5 s after the appearance of a new target. If the patient does not start the movement before this time, the robot's stiffness begins with



(a) DP



(b) IE

Fig. 6. Relationships between $\hat{K}_{h-m}$ and (left) T_r , (right) V_m parameters for (a) DP and (b) IE movements. Graphs show the moving average ($n = 101$) of the scatter plots. Horizontal axis represents the values for mean estimated patient's stiffness, and vertical axis represents the parameters.

the maximum allowed value and the robot starts the movement. Robot stiffness decreases according to the increase of patient's participation following the equations (6), (10), and (11). As predicted in the Section II-C, the optimal method showed a higher decrease rate than the complementary one.

Estimation and assistance period ends when the position error is lower than θ_e^{adm} , and the robot's stiffness gradually decreases to zero. Once the target is reached, it is given a time of about 0.8 s for the appearance of a new target.

B. Relationships between $\hat{K}_h$ and kinematic parameters

In order to determine the feasibility of the method used for estimation of the patient participation, we established quantitative relations between the estimated stiffness of the patient and the parameter values of the movements. Initially, we computed the average of the patient's stiffness during the assistance period:

$$\hat{K}_{h-m} = \frac{1}{T_a - T_e} \sum_{n=T_e}^{T_a} \hat{K}_h(n). \quad (18)$$

where T_e is the time when starts the stiffness estimation.

It was observed that there is a significant inverse correlation between time T_i and the mean estimated patient's stiffness, $R^2 = 0.26$, $p < 0.05$. However, such correlation was no observed in IE movements, $R^2 \approx 0$, $p < 0.05$, possibly explained by the shorter time before the start of the robot assistance. Statistical correlations were not found in the case of T_r parameter, $R^2 \approx 0$, $p < 0.05$, however, Figures 6 show that the increase of $\hat{K}_{h-m}$ might be associated with the T_r reduction, especially in higher values of the stiffness.

Correlation was also observed between $\hat{K}_{h-m}$ and the mean velocity during a DP movement. From Figure 6 is

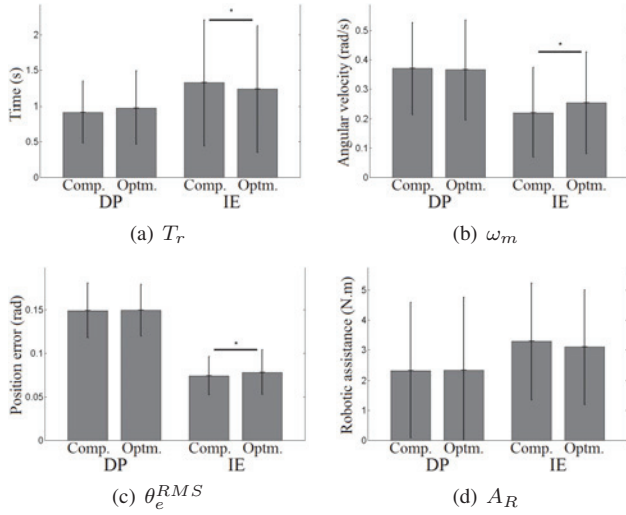


Fig. 7. Comparison between complementary and optimal methods. Mean $\pm$ SD for parameters (a) T_r , (b) ω_m , (c) θ_e^{RMS} and (d) A_R , comparing the results for the methods of assistance for DP and IE movements. Bars marked with an asterisk means that the data are significantly different ($p < 0.05$ for non-parametric Wilcoxon-Mann-Whitney test).

possible to identify a increasing trend in the values of the ω_m value with the increasing of $\hat{K}_{h-m}$, $R^2 = 0.10$, $p < 0.05$. For IE movements, increase of the value of the speed parameter was observed only in higher values of the stiffness, Figure 6. It is expected that inclusion of damping estimation will improve velocity-dependent correlations.

C. Comparison between assistance methods

We established quantitative differences between the results for the two methods used to compute the robot's stiffness (complementary and optimal), Figure 7.

For DP movements, we did not observe significant differences for the values of the different parameters between the two methods of assistance. Initially, this result refutes our previous observations [9], in which the optimal method induces some characteristics of better patient recovery, including less time to complete the task and higher velocities. However, these behaviors were observed for IE movements.

We observed that the time T_r was significantly lower using the optimal method, 1.23 ± 0.88 s compared to 1.32 ± 0.88 s, $p < 0.05$, Fig. 7(a). Also, the average velocity was significantly higher when we used optimal assistance, 0.25 ± 0.17 m/s compared to 0.22 ± 0.15 m/s, $p < 0.05$, Fig. 7(b). A higher average power during the movement, shown in a shorter time, and a greater average velocity would indicate more effort on the part of the patient either by his/her increasing the muscular capacity and/or participation.

Regarding the comparison between the mean quadratic error values during the movements, we found that the error is significantly higher for the optimal case, 0.078 ± 0.025 m/s compared to 0.074 ± 0.021 m/s, $p < 0.05$, Fig. 7(c). Hence, we can assert that the optimal method allows more error during the movement.

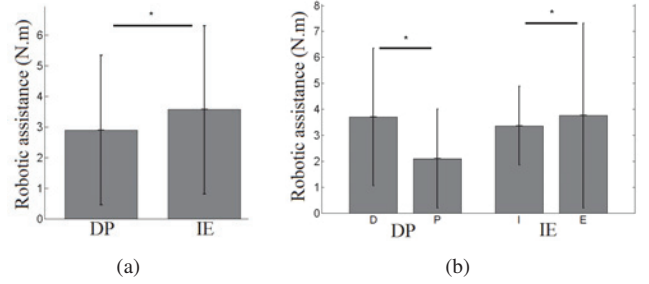


Fig. 8. Comparison of robotic assistance between directions of movement. Mean $\pm$ SD for (a) DP and IE movements and (b) Dorsiflexion, Plantarflexion, Inversion and Eversion movements.

Finally, we did not find significant differences between the amount of robotic assistance provided using the two methods, Fig. 7(d), $p < 0.05$.

D. Comparison of robotic assistance between DP and IE.

The proposed adaptive strategy aims to provide a higher amount of assistance when the patient was unable to perform the movement. The previous results as well as the feedback provided by the patient allow us to identify that the eversion/inversion movements were much more difficult to perform than dorsi/plantarflexion movements. This is also evidenced by the greater amount of assistance required for movements of this axis, Figure 8(a).

Significant differences were observed between the amount of assistance provided during movements of dorsal and plantar flexion. To recognize the importance of this difference, we must take into account that the gravitational effects were neglected in the algorithm formulation. The amount of assistance provided to dorsal flexion (against to gravity) was much higher than for plantar flexion, Figure 8(b). Thus, this result demonstrates that the methodology provides adaptation, since the robot provides less support to the patient for less difficult movements.

E. Results of the performance-based adaptation

According to the proposed performance-based adaptation, if the patient has a good performance, the robot assistance decreases due to the forgetting factor f . But, if the patient has a poor performance, the robot assistance increases due to the gain factor g . Patients sustained a high performance during the tests, averaging 89.2% of targets achieved. Besides the benefits to the patient's motivation, this result allows us to recognize that the robot assistance was helpful to the patient, despite the slight variations between the methods to determine K_r . In general, the good performance of patients during the evaluations allowed the level of robot assistance, represented by α , to decrease significantly (from 0.5 to 0.32 at best), decreasing the maximum instantaneous robotic assistance into 35%, Figure 9.

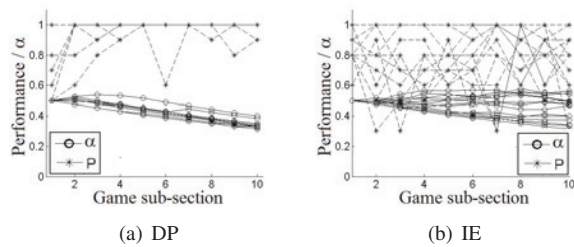


Fig. 9. Performance-based adaptation. Graphs show α and P for DP (left) and IE (right). Each line corresponds to one trial.

IV. DISCUSSION

In this study, the results observed for the ankle movements allow us to validate the online variation of the robotic assistance according to the patient's instantaneous participation, and its progressive adaptation according to the patient's performance over each therapy session. A key aspect of the results lies in the validity of the methodology used to estimate the stiffness of the patient as a measure of patient's instantaneous participation. If we define a reference position for a movement, the strategy instantly determines the amount of assistance needed by the patient from the position error value and the amount of assistance that the robot is currently providing. One consequence of this formulation is that the system identifies whether the current position was achieved by patient effort or by the robot action, and uses this information to estimate the patient's participation. This formulation, unlike previous adaptive strategies, changes the amount of assistance by modifying the parameters of the controller for each instant of time rather than for each complete movement or for each phase of gait [12], [13].

In this work, the concept of estimated stiffness of the patient corresponds to the zero order component of the dynamic response of the ankle joint, calculated for a given reference position and under the effect of a given constant external force. Although the methodology used does not allow a complete or accurate characterization of the ankle impedance, the order of magnitude observed in the results (10 – 100 N.m/rad) corresponds with that observed in more complex studies of characterization of the ankle impedance [10], [14]. Furthermore, the quantitative relationships observed between the estimated stiffness and some parameters, such as time and speed, confirm that the proposed estimation method provides relevant information about the patient.

The results presented in this paper show some limitations. First, the small number of patients involved in the assessment limits their generalization. However, this paper was a first step in evaluating the new method, and as a test of usability, was able to identify the requirements necessary for use in more complete studies. To evaluate the efficiency of the proposed strategy, future studies must be performed involving larger groups, randomly selected, and using a control group receiving traditional therapy or robotic devices with no adaptive stiffness. Second, as observed in Section III-B, to

improve the analysis of the velocity-dependent correlations, the characterization of the patient's impedance must include the higher order components such as damping and inertia. Third, since we define a desired final position instead of a desired trajectory, the error will be always large at the beginning of each movement, resulting in a low value of estimated stiffness, independently of the patient effort.

V. CONCLUSIONS

In this paper, we proposed and evaluated two adaptive impedance control schemes for rehabilitation robotics, with experimental validation using the Anklebot. The two approaches to determine the robot's stiffness, complementary and optimal, were based on online estimation of the active participation of the patient and on his/her performance playing a video game. Experimental results show that the proposed control strategies can effectively estimate the patient's participation and properly set the level of robot assistance in order to complete the task.

REFERENCES

- [1] W. H. O. WHO, *Atlas of Heart Disease and Stroke*. Geneva: WHO Press, 2011.
- [2] R. Riener, L. Lunenburger, S. Jezernik, M. Anderschitz, G. Colombo, and V. Dietz, "Patient-cooperative strategies for robot-aided treadmill training: first experimental results," *IEEE Trans. Neural Syst. Rehab. Eng.*, vol. 13, no. 3, pp. 380–394, Sept 2005.
- [3] M. Zhang, D. T. Claire, and S. Xie, "Effectiveness of robot-assisted therapy on ankle rehabilitation – a systematic review," *Journal of NeuroEngineering and Rehabilitation*, vol. 10, no. 1, 2013.
- [4] H. I. Krebs, J. J. Palazzolo, L. Dipietro, M. Ferraro, J. Krol, K. Rankeleiv, B. T. Volpe, and N. Hogan, "Rehabilitation robotics: Performance-based progressive robot-assisted therapy," *Autonomous Robots*, vol. 15, pp. 7–20, 2003.
- [5] L. Marchal-Crespo and D. Reinkensmeyer, "Review of control strategies for robotic movement training after neurologic injury," *Journal of NeuroEngineering and Rehabilitation*, vol. 6, no. 1, p. 20, 2009.
- [6] K. P. Michmizos and H. I. Krebs, "Assist-as-needed in lower extremity robotic therapy for children with cerebral palsy," in *International Conference on Biomedical Robotics and Biomechanics*, 4, 2012. Rome: IEEE, 2012, pp. 1081–1086.
- [7] N. Hogan, "Impedance control: an approach to manipulation," *Journal of Dynamic Systems, Measurement and Control*, vol. 107 (Parts 1-3), no. 1, pp. 1–24, 1985.
- [8] S. Jezernik, G. Colombo, and M. Morari, "Automatic gait-pattern adaptation algorithms for rehabilitation with a 4-dof robotic orthosis," *IEEE Trans. Robotics Autom.*, vol. 20, no. 3, pp. 574–582, June 2004.
- [9] J. C. Ibarra, W. M. dos Santos, H. I. Krebs, and A. A. Siqueira, "Adaptive impedance control for robot-aided rehabilitation of ankle movements," in *5th IEEE RAS EMBS International Conference on Biomedical Robotics and Biomechanics*, Aug 2014, pp. 664–669.
- [10] A. Roy, H. I. Krebs, D. J. Williams, C. T. Bever, L. W. Forrester, R. M. Macko, and N. Hogan, "Robot-aided neurorehabilitation: A novel robot for ankle rehabilitation," *IEEE Trans. Robotics*, vol. 25, no. 3, pp. 569–582, June 2009.
- [11] N. Jarrassé, T. Charalambous, and E. Burdet, "A framework to describe, analyze and generate interactive motor behaviors," *PLOS ONE*, vol. 7, no. 11, p. e49945, 2012.
- [12] J. A. Blaya and H. Herr, "Adaptive control of a variable-impedance ankle-foot orthosis to assist drop-foot gait," *IEEE Trans. Neural Syst. Rehab. Eng.*, vol. 12, no. 1, pp. 24–31, March 2004.
- [13] J. L. Emken, J. E. Bobrow, and D. J. Reinkensmeyer, "Robotic movement training as an optimization problem: designing a controller that assists only as needed," in *International Conference on Rehabilitation Robotics*, 9, 2005. Chicago: IEEE, June 2005, pp. 307–312.
- [14] H. Lee, "Quantitative characterization of multi-variable human ankle mechanical impedance," Ph.D. dissertation, MIT, 2013.